Visual Question Answering — Project Roadmap

MECHENG 4AI3 · Applied Artificial Intelligence · 4 Week Sprint · VQA v2.0 Dataset

Team Roles & Ownership

Student	Role	Primary Focus	Main Deliverables
A	NLP Lead 1	Question encoding, LSTM, classifier	Question encoder, answer vocab, eval metrics
B	NLP Lead 2	Embeddings, fusion, training loop	GloVe/TF-IDF, fusion module, train/eval loop
C	CV Lead 1	CNN / ResNet, DataLoader, PyTorch setup	Image encoder, VQADataset, DataLoader
D	CV Lead 2	Feature extraction, figures, repo	CNN baseline, ablations, report visuals, README

Baseline Architecture — VQA v2.0

Module	Options	Output Dim	Owner
Image Encoder	ResNet-50 (primary), VGG16, Custom CNN	2048-d	C, D
Question Encoder	BiLSTM (primary), GloVe mean-pool, TF-IDF	1024-d	A, B
Fusion Module	Concatenation or Element-wise (Hadamard)	3072-d / 1024-d	B, C
Classifier	2-layer MLP + Softmax	100-class	A
Loss	Cross-Entropy (multi-class)	Scalar	B

Task: Given image I and question Q, predict $a^* = \text{argmax}_{a \in A} P(a \mid I, Q; \theta)$ where A = top-100 most frequent answers in the training set.

Training Hyperparameters

Optimiser	Batch	Epochs	Dropout	Grad Clip	LR Scheduler
Adam lr=3e-4 wd=1e-5	64	30 (patience=5)	0.3	max norm 5	ReduceLROnPlateau factor=0.5, pat=3

WEEK 1 — Data Exploration & Preprocessing

A — NLP Lead 1

- Set up shared GitHub repo: folder structure, config.yaml, requirements.txt.
- Explore VQA v2.0: analyse question types, answer distributions, vocabulary size.
- Build question cleaning pipeline: lowercasing, tokenisation, vocabulary construction.
- Compute top-100 answer histogram; save answer↔index mappings as JSON.

B — NLP Lead 2

- Implement TF-IDF baseline vectoriser; validate on sample VQA questions.
- Load 100-d GloVe vectors; build embedding matrix; handle OOV with random init.
- Write QuestionDataset (torch.utils.data.Dataset) → (question_tensor, answer_label).

C — CV Lead 1

- Set up PyTorch environment; verify GPU / CUDA access.
- Explore VQA v2.0 image splits: inspect resolutions and diversity.
- Write image preprocessing pipeline: resize 224×224, ImageNet normalisation.
- Write ImageDataset returning (image_tensor, image_id); test batch of 64.

D — CV Lead 2

- Load ResNet-50 (torchvision); strip final FC; verify output shape (B, 2048).
- Implement custom CNN baseline: 3 conv blocks + global average pooling.
- Document output shapes; share with NLP sub-team for fusion design.

DELIVERABLES

✓ A+B: Question cleaning pipeline + QuestionDataset passing shape tests.
✓ C: Image preprocessing + ImageDataset + sample batch verified.
✓ D: ResNet-50 extractor + Custom CNN skeleton.
✓ All: Agreed tensor interface spec (image_feat_dim, question_feat_dim, batch format).

■ SYNC

NLP & CV sub-teams share output tensor shapes. A confirms ImageDataset interface with C; C confirms vocabulary label alignment with A.

WEEK 2 — Encoder Implementation & First Full Forward Pass

A — NLP Lead 1

- Implement LSTM encoder: Embedding → BiLSTM (2 layers) → 1024-d output.
- Implement VQA soft-scoring metric: acc(a) = min(count(a)/3, 1).
- Contribute to unified VQADataset merging ImageDataset + QuestionDataset.

B — NLP Lead 2

- Finalise GloVe pipeline: mean-pool token vectors → sentence representation.
- Implement ConcatFusion and HadamardFusion as swappable nn.Module subclasses.
- Write training loop: Adam, LR scheduler, early stopping, epoch-level logging.

C — CV Lead 1

- Add VGG16 as second encoder option; compare 4096-d features vs ResNet-50.
- Wire unified VQADataset; verify DataLoader shapes: (B,3,224,224)+(B,14)+(B,).
- Profile DataLoader throughput (batches/sec, GPU utilisation); tune num_workers.

D — CV Lead 2

- Implement Classifier head: Linear→ReLU→Dropout(0.3)→Linear (100 classes).
- Assemble VQAModel(image_encoder, question_encoder, fusion, classifier).
- Run first full forward pass on one batch; confirm no shape errors; profile GPU memory.

DELIVERABLES

✓ A: LSTM encoder + VQA soft-scoring metric tested.
✓ B: GloVe pipeline + ConcatFusion + HadamardFusion + training loop skeleton.
✓ C: Unified VQADataset + DataLoader throughput benchmark.
✓ D: Classifier head + assembled VQAModel + full forward pass confirmed.

■ SYNC

All members run the full forward pass together. Verify: loss computes, backward() runs without NaN, all encoder combos produce valid logits. Fix mismatches before Week 3 training.

WEEK 3 — Model Training & Ablation Experiments

A — NLP Lead 1

- Implement evaluation pipeline: Precision, Recall, F1 (macro + per-class) via sklearn.
- NLP ablation: TF-IDF vs. GloVe vs. LSTM, holding ResNet-50 + Concat fixed.
- Error analysis by question type: bin into yes/no, counting, colour, spatial.
- Draft report Section 3 (question encoder design, vocabulary selection).

B — NLP Lead 2

- First full training run: ResNet-50 + LSTM + ConcatFusion, 20 epochs; log to W&B;/CSV.
- Ablation: Concat vs. Hadamard fusion, ResNet-50 + LSTM fixed; report F1 for both.
- Implement attention-weighted GloVe mean-pool as one enhancement over baseline.

C — CV Lead 1

- CV ablation: ResNet-50 vs. VGG16 vs. Custom CNN, LSTM + Concat fixed; log all results.
- Fine-tuning experiment: unfreeze ResNet-50 layer4 vs. fully frozen; compare val accuracy.
- Produce training curves: loss/accuracy vs. epoch (train + val) for ≥ 3 configurations.

D — CV Lead 2

- Spatial features: extract 7×7 grid from ResNet-50 layer4 (2048×7×7); compare vs. global pool.
- Build confusion matrix for top-20 answer classes; identify most confused pairs.
- Begin qualitative gallery: 15 examples (successes + ≥5 clear failures) with annotations.

DELIVERABLES

- ✓ A: Precision/Recall/F1 pipeline + question-type error analysis.
- ✓ B: Full training run logged + Concat vs. Hadamard ablation results.
- ✓ C: CV ablation table (ResNet vs. VGG16 vs. CNN) + fine-tuning results.
- ✓ D: Spatial features comparison + confusion matrix + qualitative gallery (draft).
- ✓ All: Agreed best configuration for Week 4 final experiments.

■ SYNC

Review all experiment logs together: (1) Does LSTM beat TF-IDF by ≥3% F1? (2) Does ResNet-50 beat custom CNN by >5%? (3) Is Concat or Hadamard fusion better? Agree on final model config.

WEEK 4 — Evaluation, Report, Presentation & Submission

A — NLP Lead 1

- Full eval on VQA v2.0 val split: Precision, Recall, F1 + VQA Accuracy.
- Final comparison table: all encoder × fusion combos, 95% CI over 3 seeds.
- Write NLP sections of report: encoder design, answer classification, fusion analysis.
- Prepare 4–5 NLP slides: pipeline diagram, encoder bar chart, error analysis.

B — NLP Lead 2

- Implement simple cross-attention: question tokens attend to image spatial features.
- Compare improved model vs. baseline; report ΔF1 and ΔVQA Accuracy.
- Write report Sections 3–5: related work, methodology, results tables, ablation discussion.
- Contribute fusion strategy slides + lessons learned to presentation.

C — CV Lead 1

- Hyperparameter sweep: batch_size ∈ {32,64,128} × lr ∈ {1e-3, 3e-4, 1e-4}; find best val-F1.
- Write CV sections of report: encoder design, feature extraction, spatial vs. global pooling.
- Reproduce ≥1 reference result (Medium article or OpenCV blog) as comparison data point.

D — CV Lead 2

- Polish all report figures: consistent colours, clear axis labels, professional captions.
- Expand qualitative gallery to 25 annotated examples (successes, edge cases, failures).
- Write Introduction, Related Work, and Conclusion; proofread full document.
- Lead final integration: clean codebase, write README.md with install + quick-start.

DELIVERABLES

- ✓ A+B: Full report (12–18 pages) + complete comparison table + improvement results.
- ✓ C: Hyperparameter sweep results + final val-set evaluation.
- ✓ D: Publication-quality figures + 25-example gallery + clean GitHub repo + README.
- ✓ All: 10-slide presentation + 15-min rehearsed demo (live or pre-recorded).

■ SYNC Final submission: report submitted, all experiments logged with CIs, GitHub repo public and reproducible from a single script.

Report & Presentation Requirements

Scientific Report (12–18 pages)	Oral Presentation (10 slides · 15 min)
Section 1 — Introduction (D) Motivation, problem statement, contributions overview. Section 2 — Related Work (D) VQA datasets, CNN image encoders, NLP encoders, fusion methods. Section 3 — Methodology (A, B, C) Architecture diagram, encoder choices, fusion design, training details. Section 4 — Experiments & Results (A, B, C, D) Ablation tables, training curves, confusion matrix, qualitative gallery. Section 5 — Discussion & Conclusion (B, D) Limitations, error analysis, improvement attempt, future work.	Slide 1 — Title, team, dataset (All) Slide 2 — Problem & motivation (D) Slide 3 — Architecture pipeline diagram (C) Slide 4 — Image encoder comparison (C, D) Slide 5 — Question encoder comparison (A) Slide 6 — Fusion strategy results (B) Slide 7 — Main results table + training curves (A, B) Slide 8 — Qualitative gallery (D) Slide 9 — Improvement attempt Δ results (B) Slide 10 — Conclusion & lessons learned (All)

Common Pitfalls to Avoid

P1 — Answer vocab leakage	Build the top-100 vocabulary from the training split only — never use val/test annotations.

P2 — Normalisation mismatch	Always apply ImageNet mean/std when using pretrained ResNet or VGG16 (forgetting typically costs 10–15% accuracy).
P3 — Padding the LSTM	Pad all questions to 14 tokens before batching, or use pack_padded_sequence to avoid corrupting hidden states.
P4 — Macro vs. accuracy	Report macro-F1 alongside accuracy — predicting 'yes' always gives ~25% accuracy but near-zero F1.
P5 — Fusion dim mismatch	Hadamard fusion requires v and q to share the same dimension — add linear projections before fusing if needed.

Evaluation Checklist

■ VQA Accuracy on val split > majority-class baseline (~25%).	■ Single config.yaml controls all hyperparameters and encoder choices.
■ F1-score reported macro-averaged over all 100 answer classes.	■ No data leakage: vocabulary built from training split only.
■ LSTM encoder outperforms TF-IDF by at least 3% F1.	■ GitHub repo public; pip install -r requirements.txt reproduces the environment.
■ ResNet-50 outperforms custom CNN by more than 5% accuracy.	■ README includes: task description, quick-start command, results table.
■ Confidence intervals provided (3 random seeds minimum).	■ Qualitative gallery: 25 examples with correct predictions and failure modes.
■ At least one improvement attempt beyond the baseline architecture.	■ Presentation rehearsed to 15 minutes with a live or pre-recorded inference demo.